

Bailey Shelden
INFO 2201
Instructor Anas Buhayh
1459 Words & Works Cited

Final Project Writeup

With the November 2024 election looming over American's heads, I want to take an international look at women's rights and roles. By examining and visualizing trends in women's rights around the world across time. I aim to identify patterns and trends in areas such as gender equality, legal rights, labor force participation, violence against women, and healthcare, which will paint a broader picture of changes.

How are women's rights changing around the world? I'll be examining different categories along the way such as gender equality, legal rights, labor force participation, violence against women, and healthcare. I will be examining an World Health Organization Global Health Observatory (WHOGHO) API, and two datasets, totaling 6 CSV files. I aim to create visualizations which help us examine broader worldwide trends on the data to answer our research question.

API datasets were tricky to work with because they never spelled out the names of the countries, just used abbreviations and then spelled out the regions. For some instances of the dataset, I also need to rename columns such as TimeDim, ParentLocation and SpatialDim. I also renamed a few columns in the CSV files to increase readability. I also deployed pandas functions such as .describe(), .columns, .head(), .dropna(), etc.

```
In [121]: SDGIPVLT_rates_df.head(5)|
```

Out[121]:

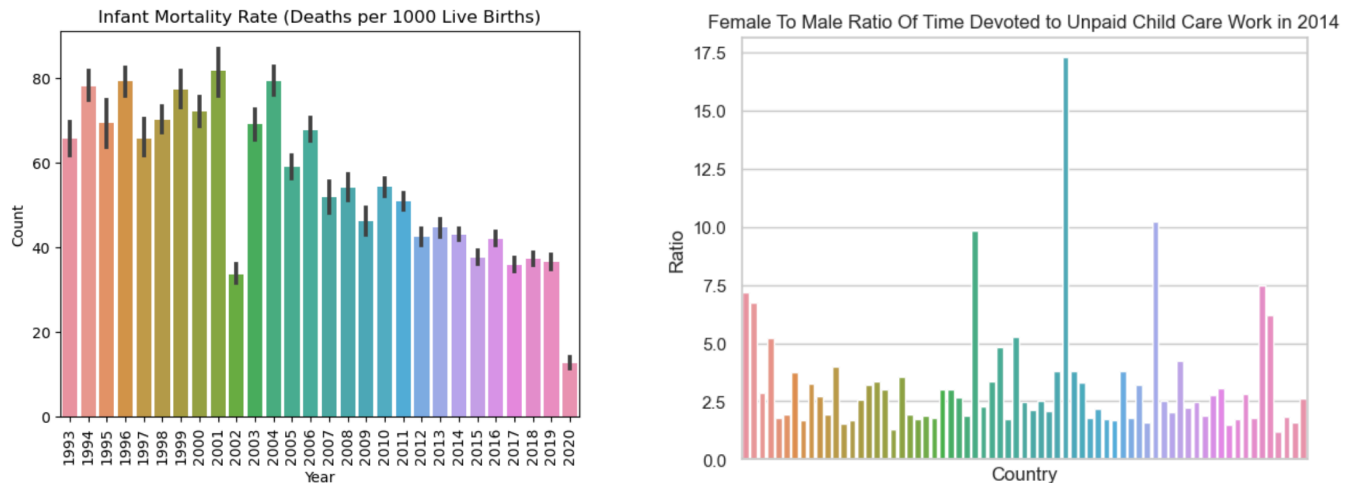
	Id	SpatialDim	ParentLocation	TimeDim	NumericValue
0	3951024	TUV	Western Pacific	2018	39.0
1	4961023	COL	Americas	2018	30.0
2	1921798	LAO	Western Pacific	2018	19.0
3	2945162	IRL	Europe	2018	16.0
4	2877132	SLE	Africa	2018	36.0

Figure 1: Image showing the result of the SDGIPVLT_rates_df first five rows in the API data frame. Here we see that 'IRL' is used to signify Ireland in SpatialDim whereas ParentLocation spells out 'Europe'

```
# Rename columns for easier access
unpaid_child_work.rename(columns={
    'Entity': 'Country',
    'Female to male ratio of time devoted to unpaid care work (OECD (2014))': 'Ratio'
}, inplace=True)
```

Figure 2: Image showing how some of the columns within CSV files needed to be renamed to increase readability and make it easier to create visualizations of those columns.

To answer the question of how each changes over time, we must examine visualizations by category: First category of **Gender Equality**, I wanted to examine time devoted to unpaid child care work and infant mortality rates. I used '1- female-to-male-ratio-of-time-devoted-to-unpaid-care-work.csv' and the WHOGHO API for these.



Figures 3 and 4: Graphs showing Infant Mortality Rate (per 1000 Live Births) and Female to Male Ratio of Time Devoted to Unpaid Child Care Word as of 2014.

In Figure 3 we are seeing a gradual decline in infant mortality since 1993. With the exception of 2002 which decreased significantly. The most valuable comparison we can grab from this graph is that we have data from 2014 specifically from several countries, 812 entries to be exact. The average infant mortality rate in 2014 according to the WHOGHO was 43. The average ratio of unpaid child work (meaning for every hour a man spends on unpaid child work, women are spending X hours) was 3.24. Again for context, that means that for every hour a man spends, a women will spend 3.24 hours on unpaid child work. From Figure 4, we can see the important distinction that around half of our countries are below 2.53. It should be noted that the lowest ratio was still not one to one, the lowest ratio belonged to Uganda with 1 to 1.18. This was surprising to me that not one country had reached one to one.

Second category of **Legal Rights**, I wanted to examine domestic violence by marital status. However, when examining the data, I quickly found a few limitations to my analysis.

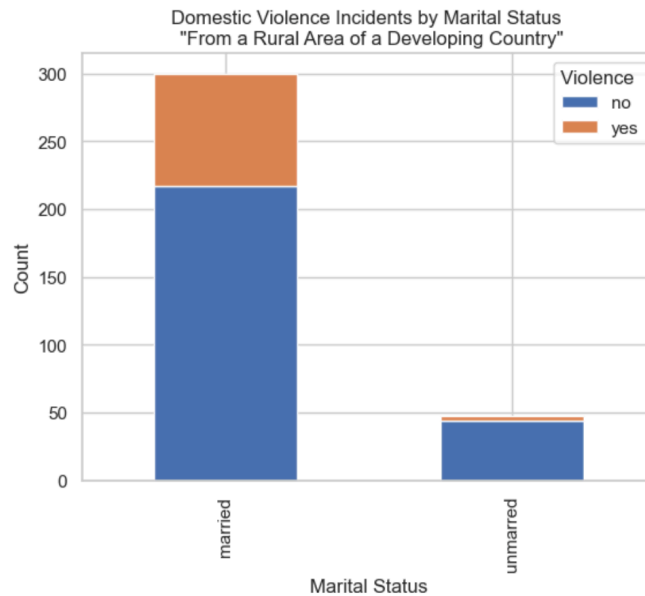


Figure 5: Plot showing Domestic Violence Incidents (self reported), which is then colored by Marital Status.

This plot was very unexpected to say the least. Before we get into the findings, we must look at the methodology of the study. This is a self reported survey of 347 women. To protect their anonymity against violent reprisals, the findings are anonymized and even some of the methodology is anonymized. The survey states it is from “a Rural Area of a Developing Country”. Looking past these caveats however, we see that it is generally unlikely for domestic violence to occur, at 24.7% percent of respondents regardless of marital status reporting violence of any kind. Only 6.3% of respondents are unmarried and experiencing domestic violence. This may shed more insight into norms and customs of this country though as 62.5% of the respondents (and the size of the married bar in Figure 5) show us a super-majority are married.

Category 3: **Labor Force Participation.** I wanted to examine labor force participation rates and the average gender wage gap. Both of these analyses will be similar to Category 1 since labor force participation rates are from 2021 whereas the gender wage gap is by country over time.

Ratio of female to male labor force participation rate (%) (modeled ILO estimate) as of 2021

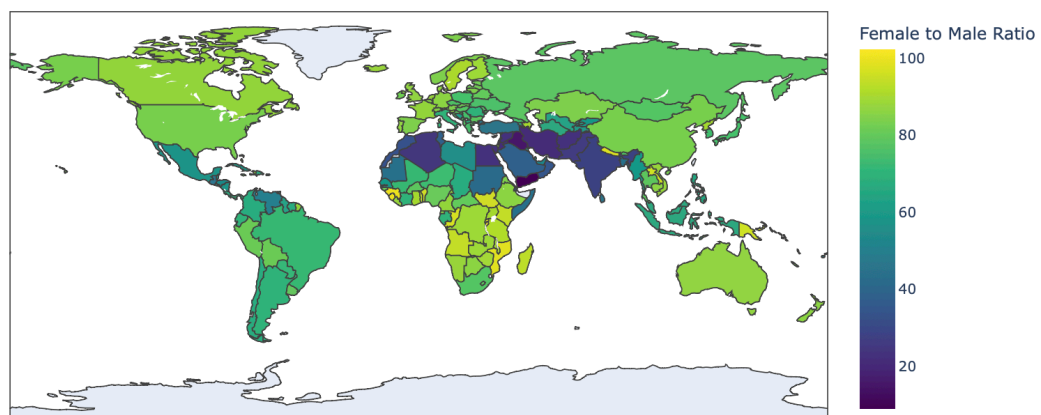


Figure 6: Choropleth showing the Ratio of Female to Male Labor Force Participation Rate as of 2021.

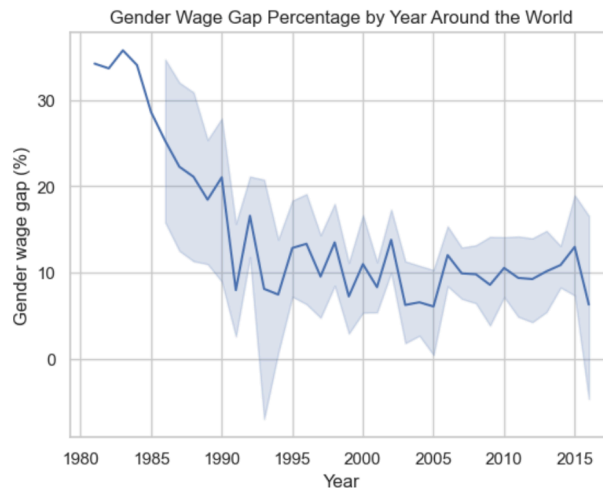


Figure 7: Line graph showing Gender Wage Gap Average Percentage by Year Around the World.

Looking at Figures 6 and 7 it is easy to see the lower levels in the ‘west’ and in 2015 (when Fig 7 ends) and think *We’ve solved feminism, let’s focus on something else*. This report is a cautionary tale, that while the green countries in Fig 6 are hovering in the 80s, they look a lot better in comparison to the blue/purple countries hovering 30s-50s. This itself is not a reason to stop working towards an 100. Every country needs to be built more, as no country is to 100. In fig 6, the coloring represents, for every 100 men participating in the labor force, X number of women are participating in the labor force. Looking at Fig 7, we are seeing a general decline from 1980 onwards to 1990 and then somewhat stalling until 2015 where we end only a little lower than the 1990 levels. Blair et al 2021¹ investigated the claim that the wage gap decreased significantly from 1980-90 and found several key factors at play such as declining unionization, reductions in gender discrimination, and reductions in education/labor market experience/occupational sorting. Economists however, are still puzzled at the stagnant wage growth post 1990. Again reference Fig 7, showing that the 2015 level is only a little lower than 1990.

Category 4: **Intimate Partner Violence**. I was not originally planning on including a section for intimate partner violence but looking through the WHOGHO endpoints, over 2800 of them, the unassuming code of SDGIVLT, proportion of women aged 15-49 who had experienced sexual and/or physical violence, caught my eye and I added a section to visualize it.

¹ Blair, Peter Q, and Benjamin Posmanick. Why Did Gender Wage Convergence Stall in the 1990s?, 13 Apr. 2021, irs.princeton.edu/document/1581.

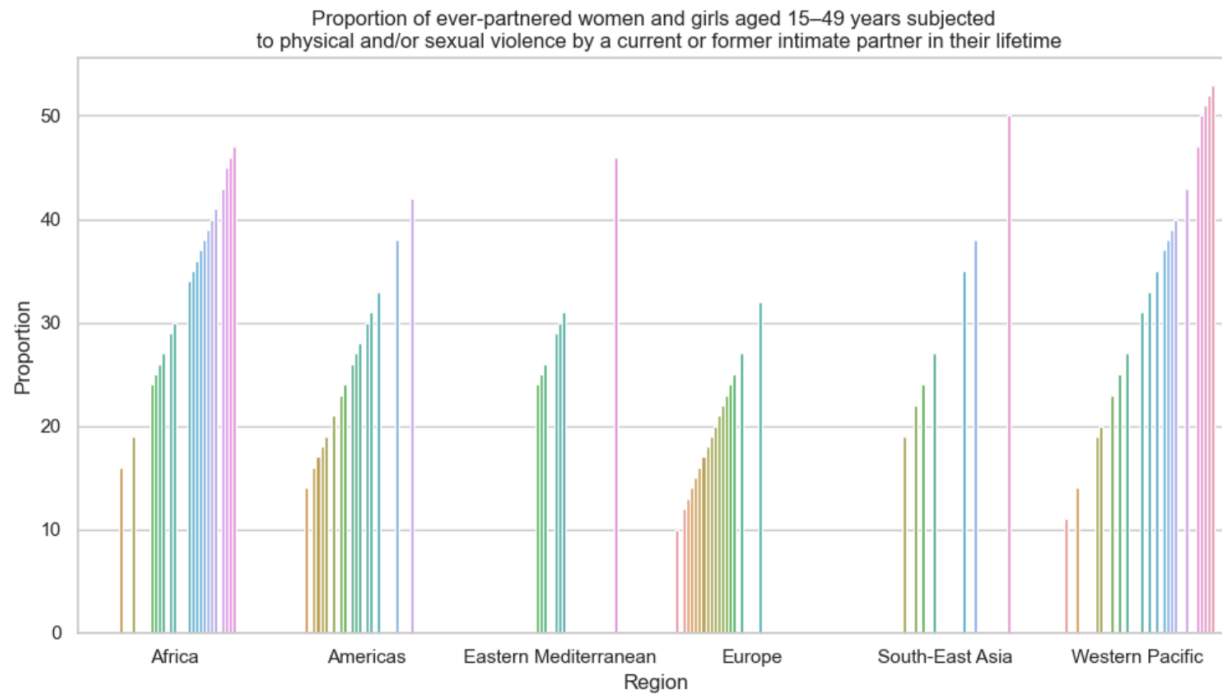


Figure 8: Graph of the Proportion of ever-partnered women and girls aged 15–49 years subjected to physical and/or sexual violence by a current or former intimate partner in their lifetime. The graph is grouped by region and each bar represents a different country. Each region shows from lowest to highest proportion.

Generating plots such as Fig 8 is interesting because the contents differ slightly if the viewer is a pessimist or an optimist. The optimist would comment on the highest bars being a little greater than 50%, which is lower than I thought it would be. The pessimist would comment that the lowest bars of Europe and Western Pacific are at or exceed 10% of the population which is still experiencing abuse. Contrary to Fig 5 of the DV Plot in a rural developing nation where the partnered abuse rate is about 24.7%, Fig 8 shows hopeful promise for developed nations where 24.7 is on the high end in each region.

Final category: **Healthcare**. Here I wanted to examine maternal mortality versus midwifery personnel around the world. Midwifery came from WHOGHO.

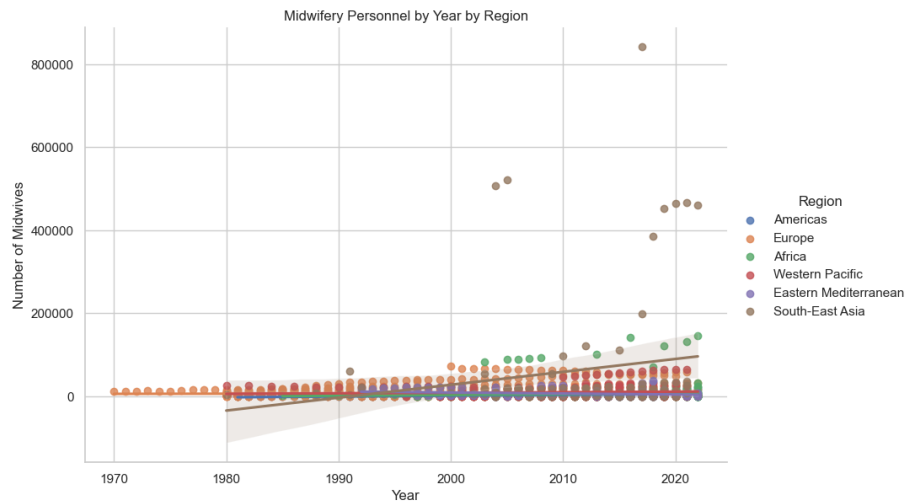
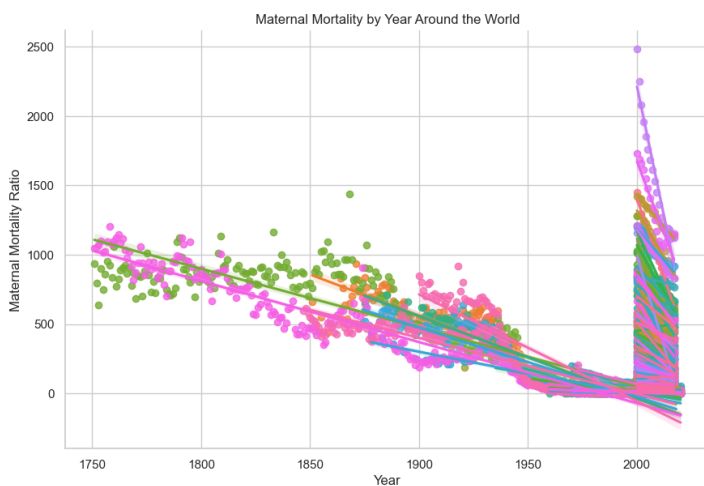
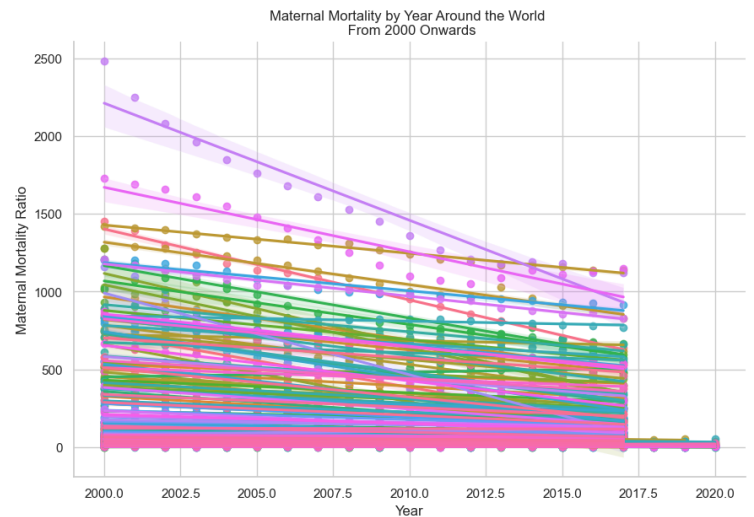


Figure 9: Linear Regression graph showing the midwifery personnel by year, colored by region. Linear Regressions are especially helpful for viewing trends over time as they create trend lines which can be positive and negative.



Source: Gapminder(2010) and World Bank(2015)



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Figures 10 and 11: Linear Regressions showing Maternal Mortality by Year Around the World. Fig 11 shows the same plot only including data from 2000 onwards.

Figure 9 shows the trend of increasing if not stagnant growth in midwifery personnel around the world. We do see a small amount of growth in the SE Asia colored trend line from 1980-2020. Plotting Fig 10 was interesting because I had no idea that there was some countries completely missing data from < 2000. Thus I wanted to make sure the trend lines stayed the same direction post 2000, hence fig 11's importance. Fig 9, 10, 11 show that since 1970, maternal mortality has been decreasing and the number of midwives globally has been slightly increasing.

Given the constraints to the API endpoints, it was hard to look at that data in a choropleth (and given that each country may have multiple entries even in the same year), it is hard to visualize the data. I would also point out that some CSV files were incomplete such as the 2- share-of-women-in-top-income-groups-atkinson- casarico-voitchovsky-2018.csv being

somewhat incomplete for entire columns or just missing many non-western countries. The domestic_violence.csv again states that the results came from a survey of 347 women in a rural area of a developing nation, to protect their anonymity. This does make it hard to generalize findings as I must rely on assumptions that the survey is unbiased and representative of that area. Given that these respondents are from a rural area, we may not be able to generalize findings to urban populations.

Again, with most of our metrics meeting their intended target, we really are seeing great strides be made in the area of women's rights. I caution making the point of this project that 'we've solved women's rights' because we have not, and it is dangerous, in terms of future progress, to quit at 90% done.

We looked at women's rights around the world over time across several key factors. We looked at gender equality through the exploration of time devoted to unpaid child care and infant mortality. We looked at legal rights through contrasting domestic violence with marriage rates. We examined labor force by looking at the share of women in top roles, labor force participation rates, and the gender wage gap. We looked at violence against women intimate partner violence stats. Finally we examined healthcare via maternal mortality and midwifery personnel numbers.

I view this work as a mid point check in to see how we are doing as a species. I think that this work highlights numerous problems in data collection but also patriarchy in a general sense of constantly prioritizing one gender over the other.

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